

# NLP For Social Media Informed Trading

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## Goal

Meme coin markets offer fast paced, extremely volatile trading opportunities with a high risk of scams. Trading these markets must be done with caution, and must be informed by social media activity, due to the lack of real value. This is why we set out to investigate relationships that exist between social media activity and asset behavior. Our two main goals include:

- Develop NLP tools that improve the trading experience
- Analyze behaviors we observe using these tools, to better characterize meme coin dynamics.

## Motivation

In recent years, social media has become a major factor in financial markets. Countless examples of assets have seen their prices soar following viral attention on platforms like Reddit and X (formerly Twitter). Though this applies to both stocks (GameStop (GME)) and cryptocurrencies (Dogecoin (DOGE)), the latter has seen a surge in popularity in recent years. Cryptocurrencies, specifically 'meme coins', are especially influenced by social media activity. These assets usually have no real value or backing, and mostly derive their value from social media marketing. Despite this seeming lack of value, these assets can be very lucrative. In fact, according to CoinGecko, the market capitalization of meme coins is estimated to be around \$60 billion dollars (April 2025), and at present forms a legitimate niche for some daytraders.

## Methods

In this project we develop the following models:

- An improved sentiment analysis model specialized for financial tweets, that assigns sentiment scores.
- An engagement model, that predicts tweet engagement metrics.
- A scam classification model that classifies whether a coin is a scam or not based on the coin's code deployed on the blockchain. (Model only available for certain blockchains)

These models enable us to look at the relationship between tweet virality, sentiment, and asset behavior. The scam models are useful in certain blockchains to protect against common exploits.

We have also developed a pipeline built on top of these models, that ingests live coins released on the blockchain, collects related tweets, and monitors their outcomes, allowing us to build a comprehensive dataset of coin behavior that we consequently analyzed.

## Sentiment Analysis

We collected data from two sources:

1. Tim Koornstra Financial Tweet Dataset
2. Abdur Rahman's Twitter Financial News Sentiment Dataset

These combined datasets provide over 50,000 annotated financial tweets. We pre-processed the tweet by replacing Twitter usernames with the placeholder token @user and URLs with the token http. Fine-tuning was performed using the Hugging Face `transformers` library and PyTorch on `cardiffnlp/twitter-roberta-base-sentiment-latest`. The model was trained for 40 epochs using the AdamW optimizer with a learning rate of  $2 \times 10^{-5}$  and a weight decay of 0.01. A linear learning rate scheduler with no warm-up steps was applied. Each training batch contained 32 examples, and gradient clipping with a max norm of 1.0 was used to prevent gradient explosion. Results can be seen in **Table 1**.

## Engagement Prediction

The engagement prediction model portion of our work aimed to predict a real value for a holistic engagement score, aggregating favorites, retweets, replies, quotes, bookmarks, and views, in hopes of capturing the virality of a tweet. We treated this as a regression problem to generate the following:

$$E(\text{score}|\text{tweet features})$$

We built our own labeled tweet dataset for tweet engagement data, since other datasets lack time information which is crucial for predicting longer term engagement stats. We trained an `xgboost` regression model whose performance can be seen in **Table 2**.

## Scam detection

For tackling the scam classification we are hoping to train a model on the source code of tokens published on the blockchain, to learn patterns between the code used and whether or not it is a scam.

$$P(\text{scam}|\text{source code})$$

**Collection Method:** We built our own simple labeled dataset for this task, that contains smart contract code, and an associated label for the ability to sell. We trained an `xgboost` classifier whose whose performance can be seen in **Table 3**

## Experiments & Results

| Model                                      | Accuracy     | F1 Score      |
|--------------------------------------------|--------------|---------------|
| <b>Fine-tuned Twitter-RoBERTa</b>          | <b>89.5%</b> | <b>0.8946</b> |
| Finetuned FinancialBERT-Sentiment-Analysis | 84.3%        | 0.843         |
| Twitter-RoBERTa                            | 57.7%        | 0.57          |
| DistilRoBERTa-Financial                    | 50.6%        | 0.56          |
| RoBERTa Base                               | 44.8%        | 0.62          |
| BERT Base Uncased                          | 32.1%        | 0.49          |
| FinTwitBERT                                | 10.0%        | —             |

Table 1: Performance comparison of sentiment classification models.

Our fine-tuned Twitter-RoBERTa surpassed every baseline, including fine-tuned FinancialBERT. This stems from training on over 50 000 labeled financial tweets—capturing Twitter-specific sentiment cues—and from targeted preprocessing that strips domain-noise like usernames, hashtags, and URLs.

| Dataset    | MSE    | RMSE   | R <sup>2</sup> |
|------------|--------|--------|----------------|
| Train      | 0.1558 | 0.3947 | 0.8413         |
| Validation | 0.1759 | 0.4194 | 0.8202         |
| Test       | 0.1690 | 0.4111 | 0.8232         |

Table 2: XGBoost model metrics

| Class        | Precision    | Recall | F <sub>1</sub> |
|--------------|--------------|--------|----------------|
| 0            | 0.42         | 1.00   | 0.59           |
| 1            | 1.00         | 0.27   | 0.42           |
| Accuracy     | 0.52 (n=102) |        |                |
| Macro avg    | 0.71         | 0.63   | 0.51           |
| Weighted avg | 0.80         | 0.52   | 0.48           |

Table 3: Test-set scam classification report

| Class        | Precision    | Recall | F <sub>1</sub> |
|--------------|--------------|--------|----------------|
| 0            | 0.27         | 0.11   | 0.16           |
| 1            | 0.63         | 0.83   | 0.72           |
| Accuracy     | 0.58 (n=102) |        |                |
| Macro avg    | 0.45         | 0.47   | 0.44           |
| Weighted avg | 0.50         | 0.58   | 0.52           |

Table 4: Honeypot.is scam classification report

Our scam prediction model outperforms online scam detection services on recall despite similar accuracy.

## Pipeline

We build a pipeline to tie all of our work together, and help us study price and social media activity dynamics. The pipeline is depicted in **Figure 1**.

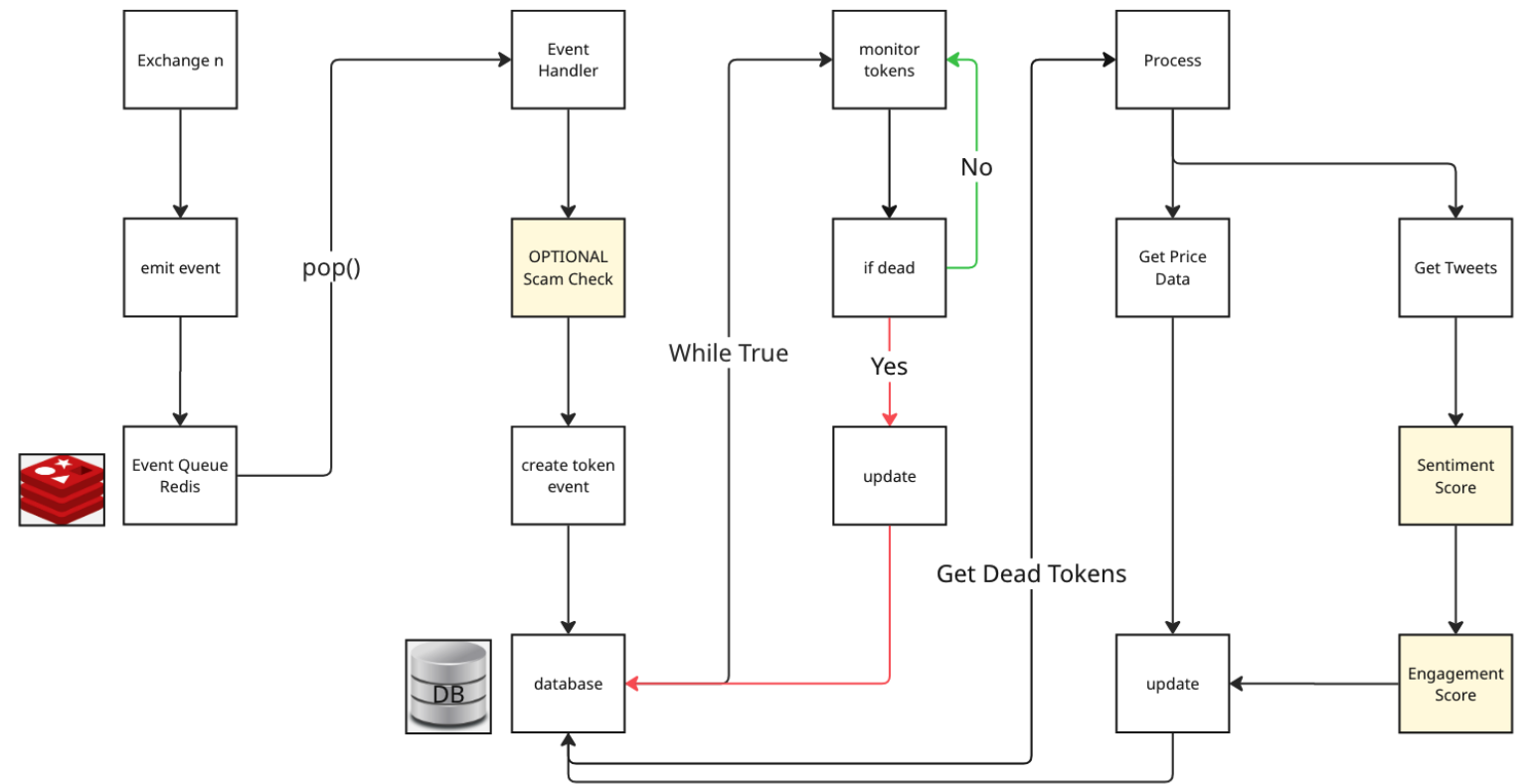


Figure 1: Pipeline Diagram.

We saw some small correlation between average tweet sentiment score, and the all time high percent price gain as shown in **Figure 2**.

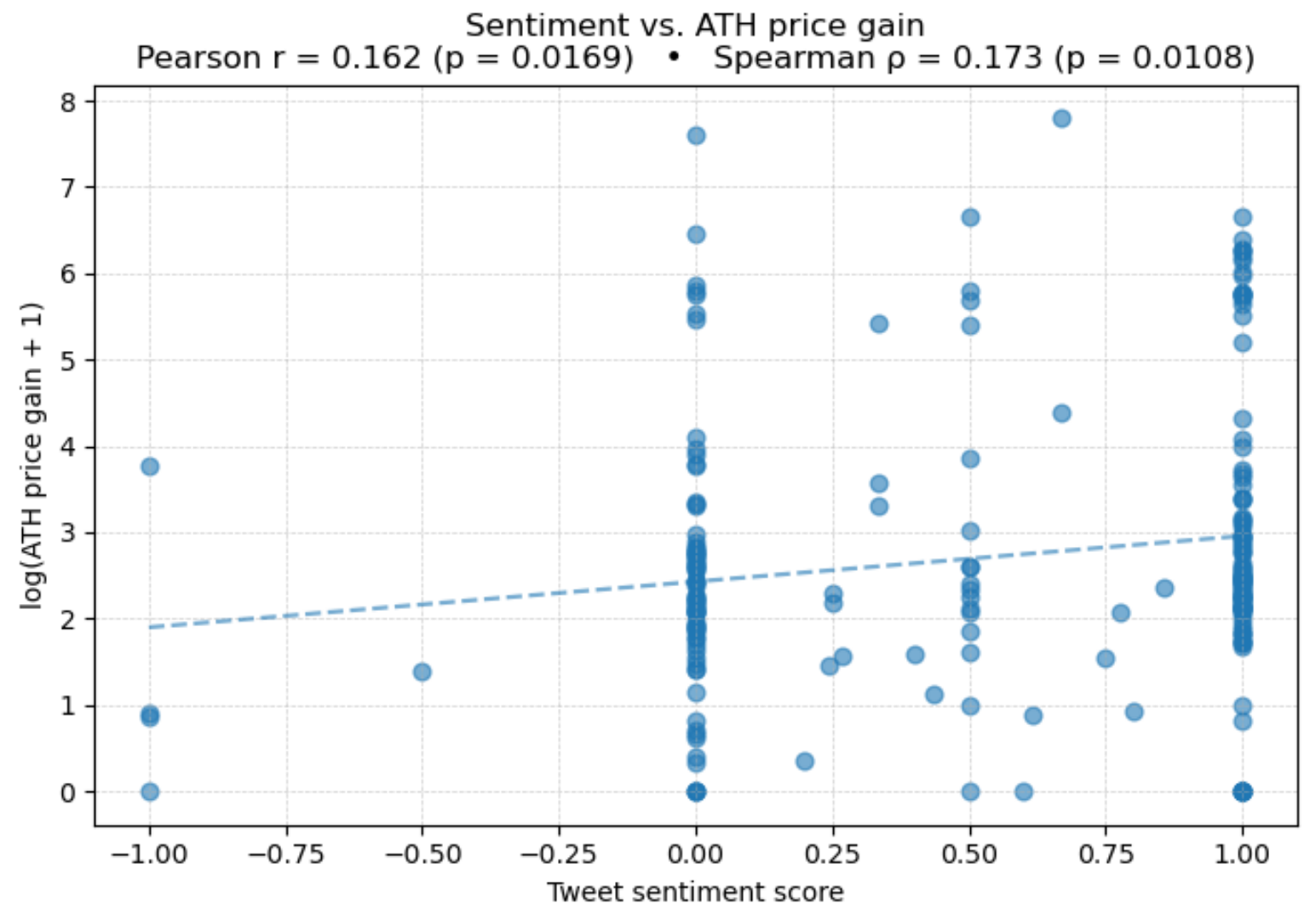


Figure 2: Sentiment vs. ATH price gain

## Conclusion

We successfully improved sentiment model performance in the context of crypto/finance tweets, we created a new model for predicting engagement with tweets, and we improved recall performance of currently available scam detection solutions. We learned so many things on this project, some notable things include:

- New scraping methods for getting data from X
- Tweet engagement behavior
- Function signatures + `tf.idf` as a powerful embedding for scam detection
- New scammer behaviors! Such as twitter account cleanup after a scam is completed
- Possible future avenues such as scammer profiling, and updating data collection pipeline